31 screens · 14 surfaces · Mobile-first, 390×844

The experience, end to end

Fourteen product surfaces, grouped by the job they do in one learning loop — not by where they sit in a menu. Every screen named here is implemented and reachable in the deployed application.

01		02		03		04		05		06		07		08		09		10
Start	→	Learn / solve	→	Submit work	→	Analyse	→	Discover gap	→	Repair	→	Practise	→	Verify	→	Track	→	Next action

THE CORE LEARNING LOOP

Traditional tools end at step 3. Every stage below is a screen a student actually passes through.

THE GAPFINDER LOOP

One mistake becomes a map

Traditional tools end at step 3. GapFinder treats a wrong answer as the *start* of the loop — and closes it only when the repair survives without help.

01 · INPUT
Work
A photo of handwriting, or typed lines. The whole attempt — not just the answer.

→

02 · AI
Reasoning
Each line is restated as a claim, so the working becomes a chain of steps.

→

03 · CODE
Verification
Every line is checked against the one above it — by algebra, not opinion.

→

04 · CODE
The gap
The *first* line that stopped following. Everything after it is a consequence.

→

05 · CODE · AI
Misconception
Named from a catalogue of 17 codes, so the same habit is countable across weeks.

→

06 · AI
Teaching
A grounded explanation, a computed diagram, and the repair worked through.

→

07 · AI · CODE
Practice
Problems built for that misconception — every wrong option is a real error.

→

08 · CODE
Verified mastery
Transfer and Exam Mode re-test with no hints. Two questions minimum, or the verdict is “uncertain”.

● **AI** — reads, narrates and phrases. Never decides. ● **CODE** — algebra decides what is true, and what mastery is worth. ● **DATA** — every stage persists its own artefact as it is computed.

The next attempt starts already knowing what broke last time

GapFinder

Diagram 02 — Core learning loop

ENTERING THE LOOP

Two doors, one destination

A student who has not written anything yet has no reasoning to diagnose. Telling them to go and make a mistake first would be a strange thing for a learning tool to do.

DOOR 1 · THEY ALREADY HAVE WORKING

Check My Work

Photograph the page, or type the lines. The pipeline reads it, restates every line as a claim, verifies each against the one above it, and stops at the **first** line that stopped following.

Ends at	A located, named, evidenced gap
If unreadable	The app says so and offers typed working — identical pipeline, no vision model
If it is only a question	No attempt to diagnose; routed to guided solving instead

DOOR 2 · THEY HAVE A BLANK PAGE

Ask a Concept · Solve With Me

A question is routed to one of 22 curated concepts and answered with a computed diagram, a read-aloud lesson and cited sources. Or the student is walked forward one move at a time.

Ends at	A concept check built from catalogued distractors
Boundary	Solve With Me names the next move and why — it never writes the line
Off-corpus	Content outside the verified library is badged differently, before it is read

Both doors converge on the same proof

A · PRACTICE

Problems built for that misconception. Every wrong option is a documented error, so choosing one names the rule the student holds.

B · TEACH-BACK

Explaining the idea back in their own words, scored against a rubric rather than a keyword match.

C · TRANSFER

The same concept inside an unfamiliar problem. Weighted double in mastery scoring, and the only event that closes a gap.

D · EXAM MODE

No hints, no feedback. Two questions minimum before any verdict, and a returning misconception is decisive.

The design consequence of the governing rule. Because correctness is decided by algebra rather than by a model, the interface can afford to be specific — it points at line 2 by name. A product that was guessing would have to hedge, and hedged feedback is the kind students learn to ignore.

GROUP 01 · DIAGNOSIS

Turning an attempt into a named gap

Three surfaces do the diagnostic work. Everything downstream in the product depends on their output being trustworthy, so all three are built to say *"I couldn't check that"* out loud.

Check My Work `/scan`

PURPOSE	Locate the first line whose reasoning stopped following.
USER ACTION	Photograph handwritten working, or type the steps. Choose the subject.
SYSTEM	Returns an analysis id immediately; the Analyzing screen polls **real pipeline position**, not a timer. One multimodal call reads the page; `mathjs` then verifies every transition and the audit locates the divergence.
VALUE	The student stops arguing with a verdict and starts looking at a line.

Analysis & Confirm `/analysis/[id]`

PURPOSE	Show the complete audit, not a boolean.
USER ACTION	Read the line-by-line verdicts; where handwriting was ambiguous, confirm what was written before anything is diagnosed.
SYSTEM	Every line carries one of five states — **correct**, **first divergence**, **downstream consequence**, **independent error**, **uncertain** — plus the corrected path, derived once and shown identically everywhere.
VALUE	Four wrong lines stop being four failures. They become the cost of one idea.

My Learning Gaps `/gaps`

PURPOSE	Make a habit visible across weeks, not just within one worksheet.
USER ACTION	Open the standing list of gaps and the error fingerprint of recurring misconception codes.
SYSTEM	Each gap carries its catalogue code and the **basis** it was established on — `proved` from an algebraic signature, or `matched` by a model from the fixed catalogue. Status moves `open → repaired → closed`.
VALUE	A stable code is countable. A model-authored label is not.

GROUP 02 · LEARNING

Teaching the one thing that broke

Every explanation is grounded in a curated corpus rather than composed freely, and each of these three surfaces has a different rule about what it may say when the corpus does not cover the question.

Ask a Concept

/learn

PURPOSE	Answer a question directly, before there is any working to diagnose.
USER ACTION	Ask in plain language — "why does the minus flip both terms?"
SYSTEM	Intent detection routes the question to one of 22 fixed concept slugs. The lesson is assembled from retrieved chunks, a **computed** diagram, and a concept check whose distractors are catalogued misconceptions.
VALUE	Choosing a wrong option is informative — it names the rule the student is holding.

Solve With Me

/solve

PURPOSE	Get a stuck student moving without handing them the answer.
USER ACTION	Enter a problem and work forward one step at a time.
SYSTEM	Names the next legal move and the reason for it, then checks the student's own line against it with the same verifier used in diagnosis. The app **never writes the line**.
VALUE	The student still performs the step, so the attempt remains theirs — and remains diagnosable.

AI Coach

/coach

PURPOSE	Answer questions about the student's own record, not just about the subject.
USER ACTION	"What do I keep getting wrong?"
SYSTEM	Retrieval-grounded and given the learner's real state — recurring codes, mastery scores, open gaps. Where the retrieved context does not cover the question, it **says so** rather than filling the space.
VALUE	Advice anchored to evidence the student can go and check for themselves.

GROUP 03 · MASTERY

Proving the repair holds without help

This group exists because a student who has just been shown the answer can reproduce it. Every surface here removes something — the hint, the familiar wording, the feedback — to find out what is left.

Practice & Teach-back

`/gaps/[id]/practice · /teach-back`

PURPOSE Repair the specific misconception, then test whether it is understood or merely copied.

SYSTEM Generated problems are **solved independently before display** — an unverified problem is never shown. Teach-back is scored against a rubric. Both are graded by the same divergence engine that made the diagnosis.

VALUE Practice moves mastery by **+6**; explaining it back sets the target from the rubric score directly.

Transfer

`/gaps/[id]/transfer`

PURPOSE Find out whether the idea survives outside the shape it was learned in.

SYSTEM The same concept placed inside an unfamiliar problem. Success is weighted **+12** against practice's **+6**, because solving an idea in a new setting is the stronger evidence.

VALUE Transfer is the **only** event that moves a gap to `closed`. Practice alone marks it `repaired`.

Exam Mode

`/exam`

PURPOSE Decide mastery under conditions where nothing is helping.

SYSTEM No hints, no feedback. Three rules in code: one right answer is never mastery (**two-question minimum**); a right answer through invalid reasoning is never mastery; a **returning misconception is decisive**, whatever the score.

VALUE Where evidence is thin the verdict is `uncertain` — a refusal to over-claim is what makes the other verdicts worth reading.

Focus Mode

`/focus`

PURPOSE Hold attention on one gap for a bounded block of time.

SYSTEM A timer bound to the live gap rather than a generic pomodoro, with optional Spotify playback through server-side OAuth (PKCE). Playback is entirely optional and never in the learning path.

VALUE The session has a subject. When it ends, there is a specific thing that was worked on.

GROUP 04 · PROGRESS & TRUST

Making the trajectory — and the system — visible

Three of these show the student their own path. The fourth shows anyone what the system actually did, which is the same commitment pointed inward.

Learning Roadmap

`/roadmap`

PURPOSE

Answer "what should I do next?" from evidence rather than a syllabus.

SYSTEM

Ordered from the seeded concept graph — 22 concepts, 29 prerequisite relationships — against current mastery, with prerequisites gated.

VALUE

The order comes from what this student's own work showed.

History

`/history`

PURPOSE

Let a student return to any past analysis intact.

SYSTEM

Every analysis by day with its outcome. Artefacts are persisted as they are computed, so **reopening a page never regenerates anything** — last week's diagnosis is the one shown today.

VALUE

Teaching that silently changes underneath a learner cannot be trusted.

Full Report

`/reports/full`

PURPOSE

Consolidate everything into one readable picture.

SYSTEM

Mastery per concept with trend and a trimmed 30-point history, recurring misconception codes, gaps by status, and outcomes across practice, transfer, teach-back and exam.

VALUE

The one screen to show a parent or a teacher.

AI Observability

`/dev/observability`

PURPOSE

Answer "what did the AI actually do here?" for any analysis.

SYSTEM

Every model call logged with stage, `provider:model`, cache status, latency, error text and the retrieved chunk ids that fed the prompt — then replayed as a per-analysis trace.

VALUE

A failover to Groq, or a fall back to deterministic logic, is visible rather than silent.

ACHIEVEMENTS · /achievements

Five seeded milestones, each with a **machine-checkable** criterion evaluated against counted evidence rather than engagement — gaps found, gaps repaired, concepts transferred, teach-backs, a seven-day streak. Nothing unlocks for opening the app.

THE FULL ROSTER — FOURTEEN SURFACES, ALL IMPLEMENTED

Check My Work · Analysis & Confirm · My Learning Gaps · Ask a Concept · Solve With Me · AI Coach · Practice & Teach-back · Transfer · Exam Mode · Focus Mode · Roadmap · History & Full Report · Achievements · AI Observability

HOW THE SURFACES CONNECT

Fourteen surfaces, one learner state

The features are not fourteen tools that happen to share a login. Each one reads and writes the same records, which is what makes a gap found on Monday the gap the roadmap orders on Tuesday.

One gap, followed through every surface that touches it

A single `Gap` row is created by **Check My Work**, listed by **My Learning Gaps**, explained on the analysis screen, drilled by **Practice**, re-tested by **Transfer**, timed by **Focus Mode**, weighted by the **Roadmap**, quoted by the **Coach**, counted toward **Achievements**, and — if its misconception code reappears — held against the verdict in **Exam Mode**. No surface keeps a private copy.

`Gap`	`MasteryRecord`	`LearningEvent`	`LearningMemory`	`Roadmap`
The diagnosis, its code, and the basis it was established on	0–100 EMA per concept, with trend and history	An append-only record of what happened, and when	Recurring gaps, successful and failed repairs, transfer results	The next-action ordering, built from the two above

WHAT THIS BUYS THE STUDENT

Nothing has to be re-explained to the app. Exam Mode already knows which habit to watch for; the Coach can cite a specific past analysis; the Roadmap reorders without being asked. The product gets more useful the longer it is used, because the evidence accumulates in one place.

WHAT IT COSTS

Every screen depends on the diagnosis being right, so a false accusation would propagate — into the roadmap, the exam, the mastery score. That is the reason correctness is decided by algebra rather than by a model, and the reason `uncertain` is a first-class outcome rather than a failure state.